

## EVENSTAD-DIH, Norway

WMO: 013830

Lat: 61.425N

Lon: 11.080E

Elev: 255

StdP: 98.30

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -25.4      | -22.6 | -27.7                         | 0.3 | -25.6 | -24.9 | 0.4 | -22.8 | 9.2                   | 1.6  | 8.4 | 0.9  | 0.9                   | 310  | 0.647 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.1                   | 26.2            | 16.6 | 24.1 | 16.2 | 21.9 | 15.0 | 18.3                | 23.2 | 17.2 | 21.6 | 16.2 | 20.2 | 2.7                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.5                            | 12.1 | 19.7 | 15.6 | 11.4 | 18.7 | 14.6 | 10.7 | 17.6 | 52.8          | 23.3 | 49.3 | 21.4 | 46.3 | 20.3 | 22.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 7.8           | 6.7  | 5.9 | DB                         | -27.9 | 28.5               | 3.6 | 2.1                                                | -30.5 | 30.0       | -32.6 | 31.2       | -34.6 | 32.4       | -37.2 | 34.0 | (4) |
| (5)               |      |     | WB                         | -29.3 | 20.0               | 3.3 | 1.4                                                | -31.7 | 21.0       | -33.6 | 21.8       | -35.5 | 22.6       | -38.0 | 23.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 3.2    | -9.1  | -7.3  | -2.4  | 3.4   | 8.6   | 12.8  | 15.4  | 13.7  | 9.7   | 3.3   | -2.5  | -7.5  | (6)  |  |
|                                                                        | DBStd       | 9.70   | 7.18  | 6.27  | 5.20  | 3.18  | 3.92  | 3.21  | 2.88  | 2.63  | 3.09  | 3.75  | 5.01  | 7.09  | (7)  |  |
|                                                                        | HDD10.0     | 2914   | 592   | 484   | 384   | 199   | 75    | 9     | 0     | 3     | 43    | 209   | 374   | 543   | (8)  |  |
|                                                                        | HDD18.3     | 5523   | 850   | 717   | 642   | 448   | 303   | 167   | 99    | 146   | 261   | 466   | 624   | 801   | (9)  |  |
|                                                                        | CDD10.0     | 443    | 0     | 0     | 0     | 1     | 30    | 94    | 168   | 117   | 32    | 2     | 0     | 0     | (10) |  |
|                                                                        | CDD18.3     | 11     | 0     | 0     | 0     | 0     | 0     | 2     | 8     | 1     | 0     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 252    | 0     | 0     | 0     | 0     | 26    | 73    | 133   | 21    | 0     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 33     | 0     | 0     | 0     | 0     | 2     | 10    | 21    | 1     | 0     | 0     | 0     | 0     | (13) |  |
|                                                                        | WSAvg       | 2.3    | 2.1   | 2.5   | 2.6   | 2.5   | 2.4   | 2.3   | 2.2   | 2.1   | 2.3   | 2.2   | 2.0   | 2.2   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 776    | 56    | 38    | 33    | 43    | 69    | 78    | 93    | 103   | 72    | 72    | 68    | 52    | (15) |  |
|                                                                        | PrecMax     | 1058   | 102   | 89    | 86    | 104   | 151   | 147   | 191   | 195   | 134   | 192   | 221   | 88    | (16) |  |
|                                                                        | PrecMin     | 589    | 13    | 11    | 0     | 4     | 3     | 3     | 21    | 27    | 32    | 12    | 22    | 6     | (17) |  |
|                                                                        | PrecStd     | 107    | 25    | 19    | 19    | 24    | 36    | 33    | 38    | 36    | 28    | 40    | 42    | 22    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 5.8   | 7.2   | 11.9  | 18.3  | 26.2  | 28.1  | 29.0  | 25.6  | 20.5  | 14.5  | 10.2  | 7.3   | (19) |  |
|                                                                        |             | MCWB   | 3.4   | 4.0   | 6.4   | 10.2  | 15.0  | 16.5  | 19.0  | 16.5  | 14.4  | 10.9  | 9.2   | 5.5   | (20) |  |
|                                                                        | 2%          | DB     | 3.8   | 4.5   | 8.6   | 15.3  | 22.8  | 25.6  | 26.7  | 23.1  | 18.6  | 12.4  | 7.5   | 5.1   | (21) |  |
|                                                                        |             | MCWB   | 2.1   | 2.4   | 4.3   | 8.3   | 13.8  | 15.5  | 17.9  | 16.2  | 13.7  | 10.1  | 6.6   | 3.6   | (22) |  |
|                                                                        | 5%          | DB     | 1.8   | 2.4   | 6.5   | 12.6  | 20.0  | 22.8  | 24.6  | 21.1  | 16.9  | 10.8  | 6.0   | 3.5   | (23) |  |
|                                                                        |             | MCWB   | 0.9   | 1.2   | 2.8   | 7.0   | 12.1  | 14.7  | 17.1  | 15.4  | 12.8  | 8.9   | 5.2   | 2.2   | (24) |  |
|                                                                        | 10%         | DB     | 0.5   | 0.9   | 4.4   | 10.2  | 16.7  | 20.2  | 22.3  | 19.2  | 15.3  | 9.2   | 3.8   | 1.9   | (25) |  |
|                                                                        |             | MCWB   | -0.1  | 0.3   | 1.8   | 5.5   | 10.4  | 13.5  | 16.1  | 14.7  | 12.0  | 7.8   | 2.9   | 1.0   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 3.9   | 4.5   | 6.7   | 10.9  | 16.4  | 17.8  | 20.4  | 18.6  | 16.0  | 12.5  | 9.5   | 5.7   | (27) |  |
|                                                                        |             | MCDB   | 5.4   | 6.7   | 11.5  | 16.1  | 23.2  | 23.3  | 25.9  | 22.0  | 18.5  | 13.7  | 10.1  | 7.1   | (28) |  |
|                                                                        | 2%          | WB     | 2.4   | 2.7   | 4.6   | 9.0   | 14.6  | 16.6  | 19.1  | 17.4  | 14.9  | 10.7  | 7.0   | 3.9   | (29) |  |
|                                                                        |             | MCDB   | 3.6   | 4.2   | 8.0   | 14.3  | 21.7  | 22.7  | 24.8  | 20.8  | 17.3  | 12.1  | 7.4   | 4.9   | (30) |  |
|                                                                        | 5%          | WB     | 1.0   | 1.4   | 3.3   | 7.4   | 13.0  | 15.5  | 18.0  | 16.5  | 13.8  | 9.3   | 5.5   | 2.4   | (31) |  |
|                                                                        |             | MCDB   | 1.7   | 2.3   | 5.7   | 12.1  | 18.4  | 21.3  | 22.8  | 19.7  | 16.2  | 10.4  | 6.1   | 3.4   | (32) |  |
|                                                                        | 10%         | WB     | -0.1  | 0.3   | 2.1   | 6.0   | 11.3  | 14.4  | 16.9  | 15.6  | 12.8  | 8.2   | 3.4   | 1.1   | (33) |  |
|                                                                        |             | MCDB   | 0.5   | 1.0   | 4.2   | 9.4   | 16.1  | 19.4  | 21.2  | 18.6  | 14.9  | 9.4   | 3.9   | 1.8   | (34) |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 6.9   | 8.4   | 10.8  | 11.2  | 11.7  | 11.8  | 11.1  | 10.4  | 10.0  | 7.2   | 5.4   | 6.8   | (35) |  |
|                                                                        | 5% DB       | MCDBR  | 6.9   | 9.0   | 12.3  | 16.3  | 18.1  | 17.1  | 15.7  | 13.9  | 12.8  | 9.8   | 6.9   | 8.3   | (36) |  |
|                                                                        |             | MCWBR  | 6.2   | 7.7   | 8.0   | 9.7   | 9.3   | 8.5   | 8.1   | 7.0   | 7.3   | 6.6   | 5.7   | 6.7   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 7.2   | 8.8   | 11.0  | 15.5  | 16.5  | 14.9  | 14.0  | 10.9  | 9.7   | 8.0   | 6.2   | 7.8   | (38) |  |
|                                                                        |             | MCWBR  | 6.0   | 7.0   | 7.3   | 9.4   | 8.5   | 7.5   | 7.5   | 6.3   | 6.4   | 6.3   | 5.7   | 6.7   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.220 | 0.270 | 0.302 | 0.340 | 0.350 | 0.344 | 0.361 | 0.352 | 0.318 | 0.302 | 0.237 | 0.186 | (40) |  |
|                                                                        | taud        |        | 2.112 | 2.233 | 2.224 | 2.235 | 2.405 | 2.470 | 2.433 | 2.452 | 2.529 | 2.472 | 2.195 | 1.996 | (41) |  |
|                                                                        | Ebn at Noon |        | 517   | 714   | 812   | 847   | 872   | 884   | 855   | 834   | 806   | 685   | 488   | 373   | (42) |  |
|                                                                        | Edh at Noon |        | 36    | 64    | 92    | 112   | 103   | 98    | 100   | 90    | 70    | 52    | 33    | 23    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.29  | 0.96  | 2.55  | 4.27  | 4.74  | 5.23  | 4.76  | 3.68  | 2.38  | 1.07  | 0.35  | 0.16  | (44) |  |
|                                                                        | RadStd      |        | 0.05  | 0.19  | 0.23  | 0.41  | 0.45  | 0.49  | 0.59  | 0.33  | 0.29  | 0.18  | 0.08  | 0.01  | (45) |  |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (53 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page